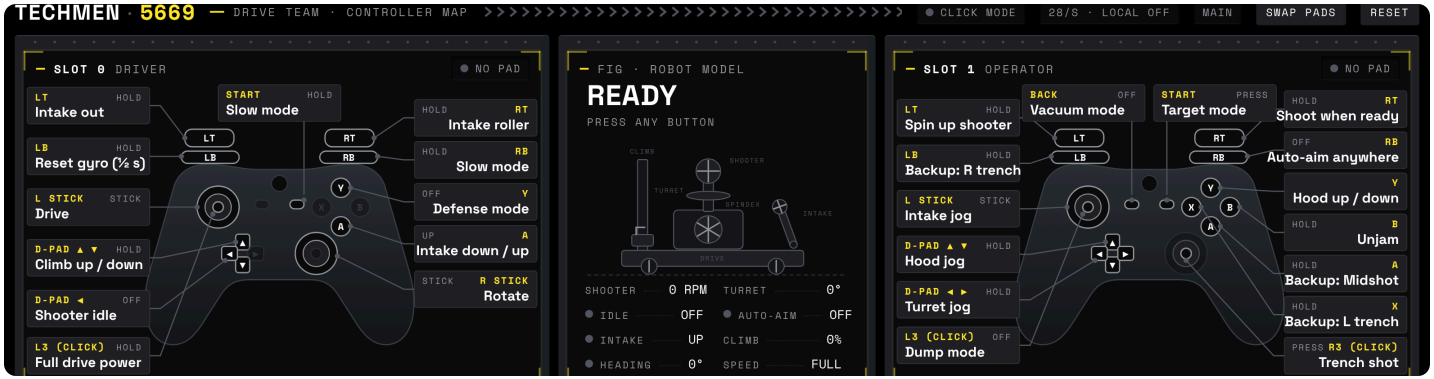
# TECHMEN 5669 · Driver & Operator Controls

Robot code: main, Oct 27 2026 · Driver = Port 0 · Operator = Port 1




## Driver · Port 0 · driving, intake, climb

| | | |
|---|---|---|
| **Left Stick** | STICK | Drive (forward, back, left, right) |
| **Right Stick** | STICK | Turn the robot |
| **RB or Start** | HOLD | Slow mode (RB keeps your thumb on the turn stick) |
| **LB** | HOLD ½ S | Reset gyro (practice only). **Robot straight away from your DS first**. Buzz = done |
| **RT** | HOLD | Intake roller: balls IN |
| **LT** | HOLD | Intake roller: balls OUT (spit them back) |
| **A** | PRESS | Intake down / up |
| **D-pad Up / Down** | HOLD | Climber up / down (stops by itself at the ends) |
| **D-pad Left** | PRESS | Shooter idle at 2000 RPM on / off |
| **Y** | PRESS | **Defense mode** on / off: intake up, hood down, quicker drive, wheels lock in an X when you let go |
| **Left Stick click** | HOLD | Full drive power in vacuum / dump mode |
| **X** | · | Nothing |

## Operator · Port 1 · aiming and shooting

| | | |
|---|---|---|
| **RB** | PRESS | **Auto-aim on / off**. It turns ON by itself at teleop start (1 buzz = on, 2 = off) |
| **RT** | HOLD | Shoot. Fires when ready (buzzes). Aim OFF: spins up, then feeds |
| **LT** | HOLD | Spin up the shooter (aim ON: RT does it itself) |
| **Y** | PRESS | Hood up / down |
| **D-pad Up / Down** | HOLD | Move the hood up / down slowly |
| **D-pad Left / Right** | HOLD | Turn the turret left / right |
| **Left Stick up / down** | HOLD | Move the intake up / down slowly (backup) |
| **B** | HOLD | Spindex backwards (unjam) |
| **A / X / LB** | HOLD ½ S | Backup shots: A = Midshot, X = Left trench, LB = Right trench |
| **Back** | PRESS | **Vacuum mode** on / off: intake + auto-aim, fires by itself (scores in our zone, passes from the neutral zone) |
| **Left Stick click** | PRESS | **Dump mode** on / off: roller OUT (X-take) + auto-pass. Intake down first |
| **Right Stick click** | PRESS | Trench shot (hood up, robot stays put) |
| **Start** | PRESS | Target: AUTO → SCORE → PASS |

Turret jog, hood jog or Y turns auto-aim OFF. Press RB to turn it back on.

# TECHMEN 5669 · How To Use It

Keep this with the driver station

## Before every match or practice

1. **Robot OFF.** Turret on the **HOME mark**. Hood **pushed all the way down** by hand. Intake **pushed all the way up**. Climber **all the way down**.
2. **Turn the robot on** while holding the hood down.
3. Driver Station **USB tab**: driver controller in **slot 0**, operator in **slot 1**. Press a button to see which row lights up; drag to fix. `F1` = rescan.
4. Driver Station **Team Station**: pick **our alliance color** (Red / Blue). At competition the field sets it.
5. Elastic: **"Pit Check"** should say **ALL GO**.
6. Practice only (a match sets it): robot **straight away from your driver station**, hold **Driver LB** until it buzzes.
7. Shop field: **"Practice Field"** switch ON. **Competition: OFF.**

## Emergency keys (Driver Station laptop)

- `Enter` = **Disable** (stops the robot, no restart needed). Use this one.
- `Space` = **Emergency stop**. The robot must be powered off and on again.

## What the buzzes mean

- **1 short** = turned ON (auto-aim, gyro reset, backup spot). **2 short** = turned OFF.
- **1 long** (both) = vacuum or dump started: the intake drops.
- Operator **steady** = ready to fire. **3 long** = hopper looks empty: go get balls.
- Operator **1 / 2 / 3 short** after Start = AUTO / SCORE / PASS.
- Driver **low hum** = our hub is open. **Pulses** = hub swap in 5 s.
- Driver **fast pulses** = pin (back off). **Steady** in the last 30 s = their tower, don't touch.
- **Both pulse** early in teleop = no FMS game data: coach sets "Auto Winner".

## Shooting with auto-aim (normal way)

1. Auto-aim turns ON by itself when teleop starts if the robot knows where it is (the turret follows our hub; the hood stays down until you shoot). Otherwise press RB.
2. Operator **holds RT**. The shooter spins up to the right speed for the distance and fires each ball the moment the shot is ready. The controller **buzzes** while it's ready.
3. While scoring, the robot slows to shooting speed and locks its wheels when you let go of the sticks. Driver left stick click = full speed. If the hub is behind the turret, the robot turns itself a little (the turn stick always wins).

## Shooting without auto-aim

1. Aim with **D-pad Left / Right** (turret) and **D-pad Up / Down** or **Y** (hood).
2. Hold **RT**: the shooter spins up and RT feeds once it is at speed (LT pre-spins).

## Backup shots (if the camera is having trouble)

1. Park on the spot: **Midshot** (between our hub and our climb), **Left trench** or **Right trench** (left / right as seen from our driver station).
2. Operator **holds A** (Midshot), **X** (Left trench) or **LB** (Right trench) for half a second (buzz = set).
3. Hold **RT**.

Live version on any driver station laptop: **tools/controller-map/Start Controller Map.bat** (lights up each button as you press it). Full details: CLAUDE.md, "Controller Bindings".